

INTERNAL GOLDEN DOCUMENTARY REFERENCE · ZERO-LOSS SYSTEM MEMORY

ELISENCE SENSOR & SIGNAL INFRASTRUCTURE

Golden Signal Provenance and Cross-Phase Architecture Reference
Authoritative Documentary Reconstruction · Level-3

Canonical Identity: ELISENCE SENSOR & SIGNAL INFRASTRUCTURE
Mission: ELISENCE_SENSOR_GOLDEN_SIGNAL_PROVENANCE_AND_CROSS_PHASE_ARCHITECTURE_REFERENCE_V2
PDF Mission: ELISENCE_SENSOR_GOLDEN_ZERO_LOSS_PDF_RENDER_V1
Classification: INTERNAL — DOCUMENTARY AUTHORITY (NOT RUNTIME PROOF)
Corpus: 22 archived files · 20 unique hashes
Census lock: 10_SENSOR_RECONCILED_CENSUS_LOCK_V2.json
Content policy: ZERO LOSS · NO SUMMARISATION · NO GOLDEN MODIFICATION

DESIGN INTENT = SUPPORTED

SENSOR-AGNOSTIC = SUPPORTED

IMPLEMENTATION = PARTIALLY_SUPPORTED

FULL RUNTIME = UNPROVEN

PHASE5 CONFLICT = YES

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ELISENCE SENSOR & SIGNAL INFRASTRUCTURE

GOLDEN SIGNAL PROVENANCE AND CROSS-PHASE ARCHITECTURE REFERENCE

AUTHORITATIVE LEVEL-3 DOCUMENTARY CONSOLIDATION (RECONCILED V2)

Field	Value
Document type	Level-3 Golden Signal Provenance & Cross-Phase Architecture Reference
Mission	ELISENCE_SENSOR_GOLDEN_SIGNAL_PROVENANCE_AND_CROSS_PHASE_ARCHITECTURE_REFERENCE_V2
Generated	2026-08-12T21:38:21+01:00
Canonical identity	ELISENCE SENSOR & SIGNAL INFRASTRUCTURE
Corpus	/SENSOR/sources — 22 archived files · 20 unique hashes
Baseline archaeology	ELISENCE_SENSOR_COMPLETE_DOCUMENTARY_ARCHAEOLOGY_AND_CROSS_PHASE_SIGNAL_RECONSTRUCTION_REPORT.md
Late-source reconciliation	ELISENCE_SENSOR_LATE_SOURCE_RECONCILIATION_AND_ARCHAEOLOGY_DELTA_REPORT_V1.md
Census lock	10_SENSOR_RECONCILED_CENSUS_LOCK_V2.json
Omission audit	11_SENSOR_RECONCILED_OMISSION_AUDIT_V2.json PASS
PDF in this mission	NO
New archaeology	NO
New source acquisition	NO

0. HOW TO READ THIS GOLDEN

This Golden consolidates already-completed archaeology. It does **not** invent runtime proof.

The Four-State Sensor Truth (do not collapse)

State	Locked assessment	Meaning
A. Architectural design intent	CROSS_PHASE_SENSOR_DESIGN_INTENT=SUPPORTED / SENSOR_INFRASTRUCTURE_DESIGN_INTENT_ASSESSMENT=SUPPORTED	ELISENCE was documentarily designed so multiple major systems/domains could operate with sensor/signal context where appropriate.
B. Sensor-agnostic architecture	SENSOR_AGNOSTIC_ARCHITECTURE_ASSESSMENT=SUPPORTED	Architecture contemplates multiple signal sources; not structurally dependent on one premium wearable vendor.
C. Implementation evidence	SENSOR_INFRASTRUCTURE_IMPLEMENTATION_ASSESSMENT=PARTIALLY_SUPPORTED	Corpus contains implementation-oriented and cross-phase evidence (e.g., MJ sensor bindings; Phase10 USAL design), but does not independently prove every intended path as completed runtime.

State	Locked assessment	Meaning
D. Full cross-ELISENCE runtime plane	FULL_SENSOR_RUNTIME_PLANE=UNPROVEN	Current documentary corpus does not independently establish complete device-to-runtime proof for the entire intended Sensor plane.

Ownership law (non-negotiable)

PHYSICAL SENSOR ≠ SIGNAL SOURCE ≠ DEVICE OWNER ≠ ACQUISITION OWNER ≠ PERMISSION OWNER ≠ INGESTION OWNER ≠ NORMALISATION OWNER ≠ QUALITY/CONFIDENCE OWNER ≠ INTERPRETATION OWNER ≠ DOMAIN INTELLIGENCE OWNER ≠ ELISA CONSUMER ≠ CHART/PROJECTION CONSUMER ≠ ALERT/PATHWAY OWNER

Maturity law

CONCEPT ≠ ROADMAP ≠ DESIGNED ≠ DOCUMENTED ≠ IMPLEMENTED_CLAIM ≠ CODE_EXPORT ≠ TESTED_CLAIM ≠ ACCEPTANCE_DEFINED ≠ PASSED_CLAIM ≠ SEALED_CLAIM ≠ GATED_CLAIM ≠ DEPLOYED_CLAIM ≠ MOUNTED_CLAIM ≠ DEVICE_PROVEN ≠ PARTIAL ≠ NOT_DONE ≠ UNKNOWN

Do **not** infer one maturity from another.

Claim-boundary law for founder / due-diligence material

DOCUMENT_STATES_X ≠ INDEPENDENT_CORPUS_CORROBORATES_X

Internal investor/due-diligence statements are documentary evidence that claims were made. They are **not** automatically independent technical proof.

1. GOLDEN FRONT MATTER — LOCKED RECONCILED CENSUS V2

Register	Locked value
ARCHIVED_FILES	22
UNIQUE_HASHES	20
SENSOR_SYSTEMS	12
SENSOR_FAMILIES	16
SIGNALS	35
CAPABILITIES	48
CAPABILITY_SOURCE_TRACEABILITY	48/48
ENGINES	13
ALGORITHMS	13
MODULES	18
INGESTION_PATHS	14
NORMALISATION_COMPONENTS	9
QUALITY_CONFIDENCE_CONTROLS	15
FUSION_ENGINES	6
CROSS_PHASE_RELATIONSHIPS	18
DOMAIN_CONSUMERS	10
ELISA_SENSOR_RELATIONSHIPS	5
SENSOR_CHART_RELATIONSHIPS	6
DOCUMENTARY_CONFLICTS	11
DOCUMENTARY_GAPS	14
DATA_STORES_REGISTRIES (baseline-preserved)	5
ROUTES_ENDPOINTS (baseline-preserved)	6

Register	Locked value
PRIVACY_PERMISSION_CONTROLS (baseline-preserved)	10
SENSOR_SAFETY_CONTROLS (baseline-preserved)	12
PROVENANCE_AUDIT_CONTROLS (baseline-preserved)	7
RESILIENCE_CONTROLS (baseline-preserved)	5
TEST_FAMILIES (baseline-preserved)	8
DEVICE_PROOF_ITEMS (baseline-preserved)	4
HISTORICAL_ERAS (baseline-preserved)	6
PHASES_REFERENCED (baseline-preserved)	8

Do **not** revert to the obsolete 17-file / 15-hash / 36-capability baseline for Golden authority.

Locked architectural assessments

Assessment	Value
SENSOR_AGNOSTIC_ARCHITECTURE_ASSESSMENT	SUPPORTED
CROSS_PHASE_SENSOR_DESIGN_INTENT	SUPPORTED
FULL_SENSOR_RUNTIME_PLANE	UNPROVEN
SENSOR_INFRASTRUCTURE_DESIGN_INTENT_ASSESSMENT	SUPPORTED
SENSOR_INFRASTRUCTURE_IMPLEMENTATION_ASSESSMENT	PARTIALLY_SUPPORTED
PHASE5_SENSOR_VS_ELISA_NUMBERING_CONFLICT_FOUND	YES
AFFORDABLE_12USD_SENSOR_STRATEGY_FOUND	YES
SCREENLESS_SENSOR_BAND_FOUND	YES
CUSTOM_FIRMWARE_FOUND	YES
STANDARD_SENSOR_TIER_FOUND	NO
PREMIUM_SENSOR_TIER_FOUND	NO
STANDARD_PREMIUM_FINAL_FEATURE_SPLIT	NOT_DOCUMENTARILY_LOCKED
CHINA_OEM_INDUSTRIALISATION_EVIDENCE_FOUND	NO
CHINA_OEM_ENGAGEMENT	FOUNDER_PROJECT_HISTORY_NOT_CORROBORATED_IN_CURRENT_SENSOR_CORPUS
MASTER_PRODUCT_ROADMAP_FOUND	YES
CONFIDENTIAL_HARDWARE_ECONOMICS_DOCUMENT_FOUND	YES

2. CANONICAL IDENTITY

Preferred architectural identity

ELISENCE SENSOR & SIGNAL INFRASTRUCTURE

A cross-cutting concern spanning acquisition, permission, ingestion, normalisation, quality/confidence, provenance, domain consumers, ELISA interaction surfaces, charts/projections, and alert/pathway edges.

Do NOT canonically define Sensor as

- permanently “Phase 5”
- merely a wearable feature
- an Apple Watch integration
- a HealthKit integration
- an AHE subsystem
- an ELISA subsystem
- a Charts subsystem

Phase-numbering conflict (preserved, not deleted)

PHASE5_SENSOR_VS_ELISA_NUMBERING_CONFLICT_FOUND=YES — Conflict ID `SENSOR-C-08`

Documentary lineage	Phase 5 meaning
Medical Journey Master Product Roadmap v1.1 (<code>Sensor-Main.pdf</code>)	Phase 5 = Sensor & Wearable Platform
Established Phase 5 ELISA archaeology	Phase 5 = ELISA
Same Master Roadmap internal wording	Also references Elisa Phase 5 for medication reminders/voice

V2 disposition class: `DIFFERENT_ROADMAP_ERA` / `PARALLEL_SUBSYSTEM_NUMBERING` / `DOCUMENT_VERSION_DRIFT`

Safe Golden rule: describe Sensor by **architectural identity**, not by a single phase number. Phase labels in filenames/roadmaps are **references**, not exclusive ownership proof.

3. FOUNDATIONAL PRINCIPLE — SENSOR IS NOT ISOLATED

Project-history / documentary design principle

ELISENCE capabilities were deliberately engineered so that, **where appropriate**, they could participate in a **common trusted signal ecosystem**.

This is design intent. It is **SUPPORTED** as documentary architecture. It is **not** a claim that every subsystem requires sensors at all times.

Precise wording (required)

Say:

ELISENCE systems were designed so that, where appropriate, they could participate in a common trusted signal ecosystem.

Do not claim:

- every line of code directly reads a sensor
- every subsystem requires sensor data
- every phase owns hardware
- every sensor path is implemented
- every signal is continuously available

Measurement ≠ Sensor

A **measurement** may originate from manual entry, clinician, caregiver, document, laboratory, device, or sensor. Sensor is one provenance class among several. Medical Journey architecture requires measurements to be **sensor-ready** without equating every measurement to a live sensor integration.

4. END-TO-END SIGNAL PROVENANCE AXIS (PRIMARY ORGANIZER)

Canonical chain for every material signal:

REAL-WORLD EVENT / PHYSIOLOGY
→ SOURCE DEVICE / SENSOR / SYSTEM
→ PERMISSION / CONSENT
→ ACQUISITION
→ INGESTION
→ NORMALISATION
→ QUALITY
→ CONFIDENCE
→ PROVENANCE / TRUST METADATA
→ DOMAIN INTERPRETATION
→ SAFETY / POLICY
→ ELISA / HUMAN INTERACTION
→ CHART / PROJECTION
→ PATHWAY / HUMAN DECISION

Stage maturity summary (documentary corpus):

Stage	Documentary state	Typical owner class
Source device / sensor	DESIGNED / PRODUCT_CONCEPT / ROADMAP platforms	Device / OEM / user device — often UNKNOWN runtime
Permission / consent	DESIGNED / DOCUMENTED (public vs consent vs clinical layers; MJ permissions)	Permission owner ≠ sensor owner
Acquisition	DESIGNED (BLE/gateway); drivers often OUT OF SCOPE	Acquisition owner
Ingestion	DESIGNED / ROADMAP / PARTIAL (API paths named; brokers/streaming often OUT OF SCOPE)	Ingestion owner
Normalisation (USAL / schemas)	DESIGNED / ROADMAP	Normalisation owner
Quality / confidence / grade	DESIGNED / APPROVED_BACKEND_AUTHORITY (MJ) / ROADMAP (Master Roadmap)	Quality owner
Provenance / Trust Layer	DOCUMENTED / DESIGNED	Trust / audit owner
Domain interpretation	ROADMAP / DESIGNED (DBL, Diabetes, AHE, Women+, Medication context)	Domain intelligence owner
Safety / policy	DESIGNED / ROADMAP (non-diagnostic; escalation ladders)	Safety / pathway owner
ELISA / human interaction	INDIRECT / CONTEXT / naming-adjacent	ELISA consumer ≠ sensor owner
Chart / projection	DOCUMENTED_BINDING / ROADMAP	Projector owner
Full device-to-decision runtime	UNPROVEN as complete plane	—

Missing stages must be marked NOT_DOCUMENTED , UNKNOWN , NOT_APPLICABLE , or CROSS_PHASE_OWNER — never invented.

5. MEDICAL JOURNEY TRUST MODEL (HIGH PROMINENCE)

Master Product Roadmap v1.1 requires every medical event — including sensor-originated events — to carry trust metadata:

Field	Purpose
Event ID	Stable identity
Source	Where it came from
Timestamp	Measurement time
Time Zone	Temporal integrity
Confidence Score	How sure the system is
Verification Status	Whether verified
Clinical Reliability	Clinical-trust framing (non-diagnostic)
Last Edited	Mutation awareness
Audit Trail	Accountability
Version History	Change lineage
Privacy Classification	Sharing/minimisation class
Data Origin	Origin taxonomy
Linked Evidence	Supporting artefacts

Why sensor data requires provenance

Users and clinicians must be able to distinguish whether information originated from:

- sensor / wearable / device
- clinician
- laboratory
- AI
- manual entry
- caregiver / other supported source

Sensor values without provenance are architecturally incomplete in the Medical Journey model.

Additional Master Roadmap sensor metadata (Phase 5 programme fields) include: `sensor_vendor`, `sensor_model`, `device_identifier`, `firmware_version`, `sdk_version`, `connection_method`, `measurement_origin`, `measurement_quality`, `measurement_confidence`, `measurement_grade`, `measurement_timestamp`, `received_timestamp`, `timezone`, `sync_session_id`, `sync_method`, `duplicate_detection_hash`.

6. MASTER MEDICAL JOURNEY ROADMAP — SENSOR ARCHITECTURE MODEL

Source authority: `Sensor-Main.pdf` — **ELISENCE MEDICAL JOURNEY PLATFORM Master Product Roadmap v1.1.**

Converted architecture model (not prose dump):

Architectural theme	Model role	Maturity in corpus
Longitudinal sensor-aware Medical Journey	Product philosophy / Capture–Connect–Understand–Act	DOCUMENTED design intent
Safe wearable integration	Platform programme	ROADMAP
Medication ↔ Sensor Data	Connect relationship	DOCUMENTED; sensor = context, not proof of intake
Sensor Correlation	Understand pillar	ROADMAP
Sensor-ready medication / adherence context	Medication execution may use sensors as context	DOCUMENTED/ROADMAP
Measurement provenance / Trust Layer	Mandatory event metadata	DOCUMENTED/DESIGNED
Sensor & Wearable Platform	Multi-vendor programme (MJ Phase5 numbering lineage)	ROADMAP
Sensor Confidence / Reliability / Conflict / Timeline	Quality & observational surfaces	ROADMAP (+ MJ APPROVED_BACKEND_AUTHORITY for related UI bindings)
Medical-grade vs consumer-grade classification	Grade control	ROADMAP
Automatic Sensor Validation	Validation control	ROADMAP
Sensor Intelligence (Phase 6H)	Domain/medical intelligence consumer of sensor context	ROADMAP
Remote monitoring (Phase 11K)	Family/Care consumer	ROADMAP
Future intelligence relationships	Forward linkage	ROADMAP/CONCEPT
Phase 4 measurements	Sensor-ready precondition; not sensor implementation	DOCUMENTED distinction

7. SENSOR SYSTEM REGISTER (12)

SEN-SYS-001 — Unified Sensor Abstraction Layer (USAL)

- MATURITY: `DESIGNED`
- SOURCE: [" `SEN-SRC-011` "]

SEN-SYS-002 — Biological Signal Bus (BSB)

- MATURITY: `DESIGNED`
- SOURCE: [" `SEN-SRC-011` "]

SEN-SYS-003 — Biological Context Engine (BCE)

- MATURITY: `DESIGNED`
- SOURCE: [" `SEN-SRC-011` "]

SEN-SYS-004 — Biological Digital Twin (BDT)

- MATURITY: `DESIGNED`
- SOURCE: [" `SEN-SRC-011` "]

SEN-SYS-005 — Digital Biomarker Lab (DBL)

- MATURITY: `ROADMAP` / `DESIGNED` / `NOT_DONE`

- SOURCE: [" SEN-SRC-014 ", " SEN-SRC-009 ", " SEN-SRC-010 ", " SEN-SRC-012 "]

SEN-SYS-006 — Emergency Escalation Engine (EEE)

- MATURITY: ROADMAP / DESIGNED / NOT_DONE
- SOURCE: [" SEN-SRC-014 ", " SEN-SRC-009 ", " SEN-SRC-010 "]

SEN-SYS-007 — Care Team Dashboard (CTD)

- MATURITY: ROADMAP / DESIGNED
- SOURCE: [" SEN-SRC-014 ", " SEN-SRC-010 "]

SEN-SYS-008 — Sensor Showcase

- MATURITY: DESIGNED; OUT_OF_SCOPE for pairing/drivers/ingestion
- SOURCE: [" SEN-SRC-011 "]

SEN-SYS-009 — Medical Journey Sensor Record Platform (UI/Backend bindings)

- MATURITY: APPROVED_BACKEND_AUTHORITY (binding docs)
- SOURCE: [" SEN-SRC-001 ", " SEN-SRC-002 ", " SEN-SRC-003 ", " SEN-SRC-004 ", " SEN-SRC-005 ", " SEN-SRC-006 ", " SEN-SRC-007 ", " SEN-SRC-008 "]

SEN-SYS-010 — Medical Journey Sensor & Wearable Platform (MJ Phase5 numbering lineage)

- MATURITY: ROADMAP
- SOURCE: Sensor-Main.pdf

SEN-SYS-011 — Medical Data Trust Layer (sensor-capable event provenance)

- MATURITY: DOCUMENTED
- SOURCE: Sensor-Main.pdf

SEN-SYS-012 — ELISENCE-owned affordable screenless sensor band hardware strategy

- MATURITY: PRODUCT_CONCEPT
- SOURCE: Confidential + Persian progress

8. SENSOR FAMILY REGISTER (16)

ID	Canonical name	Type	Maturity	Traceability
SEN-FAM-001	Heart Rate (PPG)	see archaeology	DESIGNED/ROADMAP	[" SEN-SRC-014 ", " SEN-SRC-011 ", " SEN-SRC-016 "]
SEN-FAM-002	Motion / Accelerometer	see archaeology	DESIGNED/ROADMAP	[" SEN-SRC-014 ", " SEN-SRC-009 "]
SEN-FAM-003	Skin Temperature (optional)	see archaeology	DESIGNED	[" SEN-SRC-014 ", " SEN-SRC-011 "]
SEN-FAM-004	Glucose / CGM	see archaeology	DESIGNED consumer path; optional	[" SEN-SRC-011 ", " SEN-SRC-009 ", " SEN-SRC-012 "]
SEN-FAM-005	Blood Pressure	see archaeology	DESIGNED (category)	[" SEN-SRC-011 "]
SEN-FAM-006	Oxygen Saturation	see archaeology	DESIGNED (category); roadmap notes avoid requiring advanced SpO2 hardware	[" SEN-SRC-011 ", " SEN-SRC-017 "]
SEN-FAM-007	Sleep	see archaeology	DESIGNED/OPTIONAL V1 proxy	[" SEN-SRC-011 ", " SEN-SRC-014 "]
SEN-FAM-008	Activity / Steps	see archaeology	ROADMAP	[" SEN-SRC-014 ", " SEN-SRC-009 "]
SEN-FAM-009	Tremor / Motor Instability	see archaeology	ROADMAP	[" SEN-SRC-009 ", " SEN-SRC-017 "]
SEN-FAM-010	Respiratory Pattern (from PPG)	see archaeology	ROADMAP	[" SEN-SRC-009 ", " SEN-SRC-017 "]
SEN-FAM-011	Smartwatch Ecosystems	see archaeology	DESIGNED via USAL; HealthKit named as abstracted-away dependency	[" SEN-SRC-011 "]
SEN-FAM-012	Elisence Native / Elisa Band Hardware	see archaeology	DESIGNED / STRATEGIC; NOT device-proven in this corpus	[" SEN-SRC-014 ", " SEN-SRC-016 ", " SEN-SRC-010 ", " SEN-SRC-011 "]
SEN-FAM-013	Laboratory / Body Scan / Manual Physiological	see archaeology	DESIGNED as BSB sources	[" SEN-SRC-011 "]

ID	Canonical name	Type	Maturity	Traceability
SEN-FAM-014	Camera / Phone Sensors (competitive / optional)	see archaeology	CONCEPT/COMPETITIVE (Huma) + Disney mentions mobile sensors	[" SEN-SRC-012 ", " SEN-SRC-010 "]
SEN-FAM-015	Nocturnal movement / sleep-fragmentation proxy (band/accel)	DERIVED/LOGICAL	DESIGNED/ROADMAP	Confidential + Persian DBL
SEN-FAM-016	HR variance / HRV-style variance from owned band	PHYSICAL/DERIVED	PRODUCT_CONCEPT	Confidential

Per-family ownership fields (acquisition/upstream/domain/test/device-proof) remain as reconstructed in baseline archaeology + V2 delta; no family silently upgraded to DEVICE_PROVEN.

9. SIGNAL REGISTER (35)

SIGNAL_ID	Name	RAW/DERIVED/CONTEXTUAL	Maturity	Source
SIG-001	heart_rate_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 ", " SEN-SRC-014 "]
SIG-002	resting_hr	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-003	hr_peak	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-004	hr_recovery_rate	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-005	hrv_stress_indicator	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-017 ", " SEN-SRC-016 "]
SIG-006	accelerometer_raw	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 ", " SEN-SRC-009 "]
SIG-007	step_count	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-008	sedentary_risk_score	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-009	skin_temperature	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-010	glucose_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 ", " SEN-SRC-009 "]
SIG-011	hypo_risk_score	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-009 "]
SIG-012	hypo_flag	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-009 "]
SIG-013	blood_pressure_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-014	oxygen_saturation_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-015	sleep_cycle_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 ", " SEN-SRC-014 "]
SIG-016	activity_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-017	tremor_risk_score	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-009 "]
SIG-018	gait_stability_indicator	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-009 "]
SIG-019	respiratory_pattern_from_ppg	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-009 "]
SIG-020	posture_collapse_pattern	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-009 "]
SIG-021	functional_capacity_6mwt_score	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 ", " SEN-SRC-010 "]
SIG-022	recovery_score	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-023	body_composition_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-024	hormonal_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-025	metabolic_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-026	biological_age_signal	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-027	signal_confidence	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-028	signal_quality_flags	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-029	device_trust_score	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-014 "]
SIG-030	abstract_public_indicators	see archaeology	DESIGNED/ROADMAP unless noted	[" SEN-SRC-011 "]
SIG-031	nocturnal_movement	DERIVED	DESIGNED	Confidential
SIG-032	hr_variance	DERIVED	PRODUCT_CONCEPT	Confidential

SIGNAL_ID	Name	RAW/DERIVED/CONTEXTUAL	Maturity	Source
SIG-033	measurement_grade (medical consumer)	CONTEXTUAL	ROADMAP	Sensor-Main.pdf
SIG-034	measurement_confidence	CONTEXTUAL	ROADMAP/DESIGNED	Sensor-Main.pdf + prior MJ
SIG-035	sensor_readiness (profile flag)	CONTEXTUAL	CODE_EXPORT (UI)	Sensor-2.pdf

Where documented, signals may carry: measurement timestamp, received timestamp, timezone, quality fields, confidence fields, measurement grade (medical|consumer), and provenance metadata. Undocumented units/frequencies remain `NOT_DOCUMENTED` .

10. DEVICE / PLATFORM SOURCE REGISTER

Critical law: **SUPPORTED SOURCE PLATFORM ≠ COMPLETED DEVICE INTEGRATION**

Baseline platforms (prior archaeology)

ID	Name	Maturity note
PLT-001	Elisa Band v1	see archaeology; HealthKit named ≠ implemented
PLT-002	Elisence Mobile App (Gateway)	see archaeology; HealthKit named ≠ implemented
PLT-003	Apple Watch / HealthKit (abstracted)	see archaeology; HealthKit named ≠ implemented
PLT-004	Samsung Galaxy Watch / Samsung Health APIs	see archaeology; HealthKit named ≠ implemented
PLT-005	CGM systems	see archaeology; HealthKit named ≠ implemented
PLT-006	Home BP / scales / sleep trackers / kits	see archaeology; HealthKit named ≠ implemented
PLT-007	Smartphone camera / sensors	see archaeology; HealthKit named ≠ implemented
PLT-008	Laboratory / body scan / manual entry	see archaeology; HealthKit named ≠ implemented

Additional Master Roadmap programme platforms (ROADMAP; not proof of live bridges)

ID	Name	Maturity
PLT-009	Health Connect	ROADMAP (Sensor-Main.pdf)
PLT-010	Apple Health	ROADMAP (Sensor-Main.pdf)
PLT-011	Samsung Health	ROADMAP (Sensor-Main.pdf)
PLT-012	Galaxy Ring	ROADMAP (Sensor-Main.pdf)
PLT-013	Wear OS	ROADMAP (Sensor-Main.pdf)
PLT-014	Fitbit	ROADMAP (Sensor-Main.pdf)
PLT-015	Garmin	ROADMAP (Sensor-Main.pdf)
PLT-016	Oura	ROADMAP (Sensor-Main.pdf)
PLT-017	WHOOP	ROADMAP (Sensor-Main.pdf)
PLT-018	Medical Devices (generic programme)	ROADMAP (Sensor-Main.pdf)

Also contemplated: ELISENCE-owned screenless band, manual entry, clinical/laboratory sources.

Sensor-agnostic product strategy

SENSOR_AGNOSTIC_ARCHITECTURE_ASSESSMENT=SUPPORTED

- APPLE WATCH SUPPORT ≠ APPLE WATCH REQUIREMENT
- PREMIUM WEARABLE SUPPORT ≠ PREMIUM WEARABLE REQUIREMENT
- ELISENCE SENSOR ≠ ONLY SUPPORTED SENSOR

11. AFFORDABLE SENSOR STRATEGY (MAJOR)

Locked findings:

- AFFORDABLE_12USD_SENSOR_STRATEGY_FOUND=YES
- SCREENLESS_SENSOR_BAND_FOUND=YES
- CUSTOM_FIRMWARE_FOUND=YES

Safe strategic formulation

The project explored an affordable ELISENCE-owned screenless sensor designed to provide the signal utility required by ELISENCE software without requiring every user to purchase an expensive premium wearable.

Approximate target wholesale concept: **USD 12** screenless band with custom firmware, high-accuracy HR / accelerometer, low-cost IoT framing.

Documented signal utility concept includes: HR, HR variance, step/acceleration profile, nocturnal movement, risk-pattern signatures, digital-biomarker proxy measures.

Critical \$12 claim boundary — forbidden inflations

Unless independently supported, do **not** state:

- \$12 sensor clinically equals Apple Watch
- \$12 sensor has identical accuracy to Apple Watch
- \$12 sensor replaces medical devices
- \$12 sensor has proven clinical equivalence
- \$12 sensor has mass-production proof

Price conflict preserved

SENSOR-C-09 : Elisa Band **\$12** (confidential + Persian) vs Gen1 **\$21** mention (Persian). Price not documentarily locked.

Maturity of hardware path

State	Status in current corpus
PRODUCT CONCEPT	YES
SPECIFICATION discussion	PARTIAL / commercial
MANUFACTURING DISCUSSION (documentary economics)	YES as claim document
QUOTATION	NOT_DOCUMENTED
PROTOTYPE / ENGINEERING SAMPLE / PRODUCTION / SHIPPED	NOT_DOCUMENTED

12. CONFIDENTIAL HARDWARE ECONOMICS DOCUMENT

Class: **INTERNAL FOUNDER / DUE-DILIGENCE DOCUMENT**

Title theme: Hardware Economics · OEM Strategy · Profit Model · Scale Curve

Document claim	Corroboration status	Safe Golden claim
\$12 screenless band strategy	INTERNAL_COMMERCIAL_CLAIM_ONLY (+ project-history echo)	Strategy explored; not manufacturing proof
Mfg cost \$4.2–5.0; profit \$6.5–7.8	INTERNAL_COMMERCIAL_CLAIM_ONLY	Document states X; not independently verified
200k / 1M units/month scenarios	INTERNAL_COMMERCIAL_CLAIM_ONLY	Scenario language only
90% competitor-equivalent utility	INTERNAL_COMMERCIAL_CLAIM_ONLY	Not clinical equivalence proof
60k+ LOC	INTERNAL_COMMERCIAL_CLAIM_ONLY	Not Sensor-corpus code proof
300+ tests PASS / engines operational	CONFLICTED vs archaeology	Do not treat as sealed test proof; raw logs ABSENT
Sensor-agnostic	DOCUMENTARILY_CORROBORATED	Safe
FDA CDS / non-diagnostic posture	DOCUMENTARILY_CORROBORATED as design posture	Not regulatory approval
No medical-device Class II burden	PROJECT_HISTORY_CORROBORATED as strategy	Strategy, not certification

Document claim	Corroboration status	Safe Golden claim
Near-zero opex / no clinical liability / national readiness	INTERNAL_COMMERCIAL_CLAIM_ONLY	Not established as independent fact
OEM licensing option	INTERNAL_COMMERCIAL_CLAIM_ONLY	Commercial option; no factory selected in corpus

13. STANDARD / PREMIUM PRODUCT HISTORY

Axis	Status
PROJECT_HISTORY	Standard and Premium sensor/wearable concepts existed (founder/project-history context)
CURRENT DOCUMENTARY CORPUS	STANDARD_SENSOR_TIER_FOUND=NO ; PREMIUM_SENSOR_TIER_FOUND=NO
FINAL FEATURE SPLIT	NOT_DOCUMENTARILY_LOCKED

Do not invent tier specifications. Sensor-2 “Premium*” naming is UI component branding, not a hardware tier lock (`SENSOR-C-11`).

14. CHINA / OEM HISTORY

Axis	Status
PROJECT_HISTORY	Specs prepared / China manufacturing discussions asserted outside this corpus
CURRENT SENSOR CORPUS	CHINA_OEM_INDUSTRIALISATION_EVIDENCE_FOUND=NO
Classification	FOUNDER_PROJECT_HISTORY_NOT_CORROBORATED_IN_CURRENT_SENSOR_CORPUS
Maturity floor	FOUNDER_HISTORY_ONLY

Do **not** claim factory selected, quote received, MOQ agreed, prototype ordered, production started, or shipment occurred unless future evidence proves it.

15. MEDICATION ↔ SENSOR RELATIONSHIP

Critical law:

GENERAL SENSOR SIGNAL ≠ PROOF MEDICATION WAS TAKEN

Documented model:

- Sensor data may support **adherence context**
- Sensor data must **never** be treated as diagnosis
- Sensor data never proves medication intake unless supported by approved medication-specific hardware

Contextual examples where supported: awake/activity context; device presence; future smart pillbox; CGM/vital trend context.

16. CAPABILITY LEDGER (48 / TRACEABILITY 48/48)

`SENSOR-CAP-001` — Vendor-neutral USAL normalisation

- DESCRIPTION: Normalise heterogeneous device outputs to canonical biological signals
- FAMILY: CROSS
- OWNER: `SEN-SYS-001`
- PHASE RELATIONSHIP: Phase 10 design; multi-phase enablement
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: `SEN-SYS-001`
- MATURITY: `DESIGNED`
- TEST STATE: `UNKNOWN`

- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-011 "]

SENSOR-CAP-002 — Biological Signal Bus transport

- DESCRIPTION: Transport normalised signals without interpretation
- FAMILY: CROSS
- OWNER: SEN-SYS-002
- PHASE RELATIONSHIP: Phase 10; subscribers 3/7/8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: SEN-SYS-002
- MATURITY: DESIGNED
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-011 "]

SENSOR-CAP-003 — Biological Context Engine enrichment

- DESCRIPTION: Non-diagnostic contextual enrichment
- FAMILY: CROSS
- OWNER: SEN-SYS-003
- PHASE RELATIONSHIP: Phase 10; consumers 3/7/8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: SEN-SYS-003
- MATURITY: DESIGNED
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-011 "]

SENSOR-CAP-004 — Biological Digital Twin state model

- DESCRIPTION: Longitudinal structured biological state
- FAMILY: CROSS
- OWNER: SEN-SYS-004
- PHASE RELATIONSHIP: Phase 10; readers 3/7/8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: SEN-SYS-004
- MATURITY: DESIGNED
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-011 "]

SENSOR-CAP-005 — Sensor Showcase ecosystem representation

- DESCRIPTION: Structural connected-device showcase; no drivers
- FAMILY: CROSS
- OWNER: SEN-SYS-008
- PHASE RELATIONSHIP: Phase 10
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: SEN-SYS-008
- MATURITY: DESIGNED_OUT_OF_SCOPE_RUNTIME

- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-011"]

SENSOR-CAP-006 — Public vs consent vs clinical sensor visibility

- DESCRIPTION: Layered privacy for sensor values/indicators
- FAMILY: CROSS
- OWNER: USAL/BSB/BCE
- PHASE RELATIONSHIP: Phase 10
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: USAL/BSB/BCE
- MATURITY: DESIGNED
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-011"]

SENSOR-CAP-007 — Elisa Band v1 device profile

- DESCRIPTION: HR+ACC(+optional temp), BLE, no mic/GPS/screen
- FAMILY: SEN-FAM-012
- OWNER: DeviceProfile
- PHASE RELATIONSHIP: Phase 9 roadmap / Phase 10 showcase
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DeviceProfile
- MATURITY: DESIGNED_NOT_DONE
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014"]

SENSOR-CAP-008 — Mobile gateway BLE→cloud telemetry path

- DESCRIPTION: Band→App→TLS→Cloud with offline buffer
- FAMILY: CROSS
- OWNER: Gateway
- PHASE RELATIONSHIP: Phase 9 roadmap
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: Gateway
- MATURITY: DESIGNED_NOT_DONE
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014"]

SENSOR-CAP-009 — Device pairing & telemetry APIs

- DESCRIPTION: /vX/devices/pair|telemetry|test
- FAMILY: CROSS
- OWNER: API
- PHASE RELATIONSHIP: Phase 9 roadmap
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: API

- MATURITY: DESIGNED
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014 "]

SENSOR-CAP-010 — Zero-Trust device telemetry tagging

- DESCRIPTION: source=device, device_trust_score, integrity_checks
- FAMILY: CROSS
- OWNER: Security
- PHASE RELATIONSHIP: Phase 9 roadmap
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: Security
- MATURITY: DESIGNED
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014 "]

SENSOR-CAP-011 — Digital Biomarker Lab core evaluate service

- DESCRIPTION: /vX/biomarkers/evaluate → BiomarkerResult
- FAMILY: DERIVED
- OWNER: DBL
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DBL
- MATURITY: ROADMAP_NOT_DONE
- TEST STATE: PLANNED
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014 "]

SENSOR-CAP-012 — Functional capacity 6MWT-style biomarker

- DESCRIPTION: Walk test HR+steps+accel score
- FAMILY: SEN-FAM-008
- OWNER: DBL
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DBL
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014 ", "SEN-SRC-010 "]

SENSOR-CAP-013 — Resting HR & recovery biomarker

- DESCRIPTION: Rest/activity/recovery protocol
- FAMILY: SEN-FAM-001
- OWNER: DBL
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources

- DEPENDENCIES: DBL
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014"]

SENSOR-CAP-014 — Activity regularity / sedentary pattern biomarker

- DESCRIPTION: SedentaryRiskScore
- FAMILY: SEN-FAM-008
- OWNER: DBL
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DBL
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014"]

SENSOR-CAP-015 — Sleep proxy biomarker (optional)

- DESCRIPTION: Non-diagnostic sleep fragmentation pattern
- FAMILY: SEN-FAM-007
- OWNER: DBL
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DBL
- MATURITY: OPTIONAL_ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014"]

SENSOR-CAP-016 — Tremor detection module

- DESCRIPTION: 4–7Hz ACC tremor risk score
- FAMILY: SEN-FAM-009
- OWNER: DBL add-on
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DBL add-on
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-009"]

SENSOR-CAP-017 — Gait/balance/fall-risk early signature

- DESCRIPTION: Step variance / micro-sway
- FAMILY: SEN-FAM-009
- OWNER: DBL add-on
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources

- OUTPUT: see sources
- DEPENDENCIES: DBL add-on
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-009 "]

SENSOR-CAP-018 — Respiratory pattern from PPG

- DESCRIPTION: Abnormal breathing / low perfusion flags
- FAMILY: SEN-FAM-010
- OWNER: DBL add-on
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DBL add-on
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-009 "]

SENSOR-CAP-019 — Posture collapse / prolonged inactivity alerts

- DESCRIPTION: Collapse pattern after abnormal HR
- FAMILY: SEN-FAM-002
- OWNER: DBL add-on
- PHASE RELATIONSHIP: Phase 3/8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DBL add-on
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-009 "]

SENSOR-CAP-020 — Hypoglycemia risk prediction layer

- DESCRIPTION: CGM or HR+ACC+tremor probabilistic hypo risk
- FAMILY: SEN-FAM-004
- OWNER: Phase 9 Diabetes
- PHASE RELATIONSHIP: Phase 9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: Phase 9 Diabetes
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-009 ", " SEN-SRC-017 "]

SENSOR-CAP-021 — Hypo real-time alert routing to Emergency Engine

- DESCRIPTION: Tiered vibration/audio/guardian/escalation
- FAMILY: SEN-FAM-004
- OWNER: EEE
- PHASE RELATIONSHIP: Phase 9

- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: EEE
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-009 "]

SENSOR-CAP-022 — Emergency escalation risk ladder L0-L3

- DESCRIPTION: Deterministic risk events and actions
- FAMILY: CROSS
- OWNER: EEE
- PHASE RELATIONSHIP: Phase 8/9/3
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: EEE
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-014 ", " SEN-SRC-010 "]

SENSOR-CAP-023 — Guardian/family and dialer-intent escalation

- DESCRIPTION: SMS/guardian/call recommendation pathways
- FAMILY: CROSS
- OWNER: EEE
- PHASE RELATIONSHIP: Phase 9/3
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: EEE
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-014 ", " SEN-SRC-009 "]

SENSOR-CAP-024 — Care Team non-diagnostic coaching dashboard

- DESCRIPTION: Trends + alerts for guardians/nurses/coaches
- FAMILY: CROSS
- OWNER: CTD
- PHASE RELATIONSHIP: Phase 8/10
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: CTD
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: [" SEN-SRC-014 ", " SEN-SRC-010 "]

SENSOR-CAP-025 — Unified Risk Timeline

- DESCRIPTION: Chronological DBL/Diabetes/Emergency events
- FAMILY: CROSS
- OWNER: App/Clinical

- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: App/Clinical
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-009", "SEN-SRC-017"]

SENSOR-CAP-026 — Blind/low-vision haptic+voice sensor alerts

- DESCRIPTION: Vibration patterns + spoken instructions
- FAMILY: CROSS
- OWNER: App
- PHASE RELATIONSHIP: Phase 9 add-on
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: App
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-009", "SEN-SRC-017"]

SENSOR-CAP-027 — WORM/audit evidence for sensor risk events

- DESCRIPTION: Hash-chain integrity for alerts
- FAMILY: CROSS
- OWNER: Audit
- PHASE RELATIONSHIP: Phase 0/8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: Audit
- MATURITY: DESIGNED/ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-009", "SEN-SRC-014"]

SENSOR-CAP-028 — 5-language sensor/biomarker/alert i18n

- DESCRIPTION: EN/FA/AR/TR/RO message keys + RTL
- FAMILY: CROSS
- OWNER: i18n
- PHASE RELATIONSHIP: cross-phase
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: i18n
- MATURITY: DESIGNED
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014"]

SENSOR-CAP-029 — Device fulfillment orchestration (non-ownership)

- DESCRIPTION: DeviceOrder/ShipmentStatus as orchestrator
- FAMILY: SEN-FAM-012

- OWNER: Fulfillment
- PHASE RELATIONSHIP: Phase 9 roadmap
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: Fulfillment
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014"]

SENSOR-CAP-030 — At-home lab test integration pattern

- DESCRIPTION: HomeLabTest sync to AHE/Diabetes
- FAMILY: SEN-FAM-013
- OWNER: Lab integration
- PHASE RELATIONSHIP: Phase 8/9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: Lab integration
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-014"]

SENSOR-CAP-031 — Sensor Card observational UI

- DESCRIPTION: Summarize one sensor record with quality/states
- FAMILY: CROSS
- OWNER: UI
- PHASE RELATIONSHIP: Medical Journey
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: UI
- MATURITY: DOCUMENTED
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-001"]

SENSOR-CAP-032 — Universal Sensor Record projection

- DESCRIPTION: Backend-bound sensor record read model
- FAMILY: CROSS
- OWNER: sensor_record
- PHASE RELATIONSHIP: Phase 3 module path
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: sensor_record
- MATURITY: APPROVED_BACKEND_AUTHORITY
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-003"]

SENSOR-CAP-033 — Sensor Timeline projection

- DESCRIPTION: Chronology of sensor observations

- FAMILY: CROSS
- OWNER: sensor_timeline
- PHASE RELATIONSHIP: Phase 3
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: sensor_timeline
- MATURITY: APPROVED_BACKEND_AUTHORITY
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-004"]

SENSOR-CAP-034 — Sensor validation / reliability / conflict / confidence details

- DESCRIPTION: Trust-quality UI projections
- FAMILY: CROSS
- OWNER: sensor_* layers
- PHASE RELATIONSHIP: Phase 3
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: sensor_* layers
- MATURITY: APPROVED_BACKEND_AUTHORITY
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-005", "SEN-SRC-006", "SEN-SRC-007", "SEN-SRC-008"]

SENSOR-CAP-035 — Sensor-agnostic diabetes posture (competitive gap doc)

- DESCRIPTION: Phase 9 Sensor-Agnostic claim + emergency loop gap
- FAMILY: SEN-FAM-004
- OWNER: Strategy
- PHASE RELATIONSHIP: Phase 9
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: Strategy
- MATURITY: DOCUMENTED_CLAIM
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-012"]

SENSOR-CAP-036 — Digital Biomarker Lab strategic pillar (phone+band)

- DESCRIPTION: Motion/cardio/respiratory indicators program
- FAMILY: CROSS
- OWNER: DBL
- PHASE RELATIONSHIP: Phase 7–10 strategic
- INPUT: see sources
- OUTPUT: see sources
- DEPENDENCIES: DBL
- MATURITY: IN_PROGRESS_CLAIM_DESIGN_COMPLETE
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["SEN-SRC-010"]

SENSOR-CAP-037 — Longitudinal sensor-aware Medical Journey (Master Roadmap v1.1)

- DESCRIPTION: Longitudinal sensor-aware Medical Journey (Master Roadmap v1.1)
- FAMILY: CROSS/DELTA
- OWNER: Medical Journey product architecture
- PHASE RELATIONSHIP: Cross-phase MJ Phases 0–12; Sensor programme numbered Phase 5 in this lineage
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: DOCUMENTED/ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]

SENSOR-CAP-038 — Sensor & Wearable Platform multi-vendor programme (Phase 5A–5N MJ numbering)

- DESCRIPTION: Sensor & Wearable Platform multi-vendor programme (Phase 5A–5N MJ numbering)
- FAMILY: CROSS/DELTA
- OWNER: MJ Phase5 Sensor & Wearable Platform (numbering lineage)
- PHASE RELATIONSHIP: MJ-Phase5
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]

SENSOR-CAP-039 — Medical-grade vs consumer-grade sensor classification

- DESCRIPTION: Medical-grade vs consumer-grade sensor classification
- FAMILY: CROSS/DELTA
- OWNER: Sensor Confidence / quality plane
- PHASE RELATIONSHIP: MJ-Phase5P
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]

SENSOR-CAP-040 — Automatic Sensor Validation (Master Roadmap)

- DESCRIPTION: Adjacent to prior SENSOR-CAP-034 MJ UI validation; do not collapse ownership
- FAMILY: CROSS/DELTA
- OWNER: Sensor validation plane
- PHASE RELATIONSHIP: MJ-Phase5Q
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN

- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]
- NOTES: Adjacent to prior [SENSOR-CAP-034](#) MJ UI validation; do not collapse ownership

SENSOR-CAP-041 — Screenless affordable sensor band government-wholesale economics model

- DESCRIPTION: Screenless affordable sensor band government-wholesale economics model
- FAMILY: CROSS/DELTA
- OWNER: Hardware strategy (founder due-diligence)
- PHASE RELATIONSHIP: Cross-phase hardware feeder to AHE/domains
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: [PRODUCT_CONCEPT](#) / [MANUFACTURING_DISCUSSION](#)
- TEST STATE: [UNKNOWN](#)
- SEAL STATE: [UNKNOWN](#)
- DEVICE-PROOF STATE: [NOT_DEVICE_PROVEN](#)
- SOURCE TRACEABILITY: ["سند محرمانه ساخت کارخانه سنسور" + Persian progress"]

SENSOR-CAP-042 — Custom firmware path for ELISENCE-owned sensor band

- DESCRIPTION: Custom firmware path for ELISENCE-owned sensor band
- FAMILY: CROSS/DELTA
- OWNER: Hardware/firmware (undocumented runtime owner)
- PHASE RELATIONSHIP: Hardware plane
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: [PRODUCT_CONCEPT](#)
- TEST STATE: [UNKNOWN](#)
- SEAL STATE: [UNKNOWN](#)
- DEVICE-PROOF STATE: [NOT_DEVICE_PROVEN](#)
- SOURCE TRACEABILITY: ["Confidential due-diligence"]

SENSOR-CAP-043 — Medical Data Trust Layer for sensor-originated events

- DESCRIPTION: Medical Data Trust Layer for sensor-originated events
- FAMILY: CROSS/DELTA
- OWNER: Medical Journey Trust Layer
- PHASE RELATIONSHIP: MJ foundation / cross-event
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: [DOCUMENTED/DESIGNED](#)
- TEST STATE: [UNKNOWN](#)
- SEAL STATE: [UNKNOWN](#)
- DEVICE-PROOF STATE: [NOT_DEVICE_PROVEN](#)
- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]

SENSOR-CAP-044 — Medication ↔ Sensor Data adherence context (non-diagnostic)

- DESCRIPTION: Medication ↔ Sensor Data adherence context (non-diagnostic)
- FAMILY: CROSS/DELTA
- OWNER: Medication execution / sensors as context only
- PHASE RELATIONSHIP: MJ Phase2 + sensors + Elisa Phase5 references
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: [DOCUMENTED/ROADMAP](#)
- TEST STATE: [UNKNOWN](#)

- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]

SENSOR-CAP-045 — Sensor Intelligence (Medical Intelligence Phase 6H)

- DESCRIPTION: Sensor Intelligence (Medical Intelligence Phase 6H)
- FAMILY: CROSS/DELTA
- OWNER: Medical Intelligence (not raw sensor owner)
- PHASE RELATIONSHIP: MJ-Phase6H
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]

SENSOR-CAP-046 — Family/Care remote monitoring of sensor context

- DESCRIPTION: Family/Care remote monitoring of sensor context
- FAMILY: CROSS/DELTA
- OWNER: Family & Care Network
- PHASE RELATIONSHIP: MJ-Phase11K
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: ROADMAP
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]

SENSOR-CAP-047 — OEM / firmware licensing commercial option

- DESCRIPTION: OEM / firmware licensing commercial option
- FAMILY: CROSS/DELTA
- OWNER: Commercial/OEM strategy
- PHASE RELATIONSHIP: Industrialisation (undocumented engagement)
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: INTERNAL_COMMERCIAL_CLAIM_ONLY
- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["Confidential due-diligence"]

SENSOR-CAP-048 — Sensor measurement metadata schema (vendor/model/firmware/sync/grade)

- DESCRIPTION: Sensor measurement metadata schema (vendor/model/firmware/sync/grade)
- FAMILY: CROSS/DELTA
- OWNER: Sensor Platform schema
- PHASE RELATIONSHIP: MJ-Phase5 fields
- INPUT: see late-source reconciliation
- OUTPUT: see late-source reconciliation
- DEPENDENCIES: V2 delta
- MATURITY: DOCUMENTED/ROADMAP

- TEST STATE: UNKNOWN
- SEAL STATE: UNKNOWN
- DEVICE-PROOF STATE: NOT_DEVICE_PROVEN
- SOURCE TRACEABILITY: ["Sensor-Main.pdf"]

17. ENGINE REGISTER (13)

ID	Name	Maturity	Class hint
SEN-ENG-001	USAL	DESIGNED	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-002	BSB	DESIGNED	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-003	BCE	DESIGNED	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-004	BDT	DESIGNED	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-005	Digital Biomarker Lab Core	ROADMAP/NOT_DONE	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-006	Emergency Escalation Engine	ROADMAP/NOT_DONE	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-007	Care Team Dashboard Engine	ROADMAP	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-008	Sensor Conflict Detection	APPROVED_BACKEND_AUTHORITY	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-009	Sensor Confidence Layer	APPROVED_BACKEND_AUTHORITY	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-010	Sensor Reliability Layer	APPROVED_BACKEND_AUTHORITY	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-011	Sensor Intelligence Engine (Phase 6H)	ROADMAP	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-012	Automatic Sensor Validation Engine	ROADMAP	UNKNOWN (rule/hybrid/model as per source)
SEN-ENG-013	Medical Data Trust / provenance scoring for sensor events	DOCUMENTED	UNKNOWN (rule/hybrid/model as per source)

Do not merge engines with algorithms/modules merely because names overlap. FUSION ≠ DIAGNOSIS.

18. ALGORITHM REGISTER (13)

ID	Name	Maturity
SEN-ALG-001	6MWT-style functional capacity algorithm v1	ROADMAP
SEN-ALG-002	Resting HR & recovery comparison vs age bands	ROADMAP
SEN-ALG-003	SedentaryRiskScore from movement bouts	ROADMAP
SEN-ALG-004	Sleep proxy from motion+HR	OPTIONAL_V1_ROADMAP
SEN-ALG-005	Tremor frequency-band 4–7 Hz analysis	ROADMAP
SEN-ALG-006	Gait/micro-sway fall-risk early signature	ROADMAP
SEN-ALG-007	Respiratory waveform from PPG modulation	ROADMAP
SEN-ALG-008	Collapse/prolonged inactivity after abnormal HR	ROADMAP
SEN-ALG-009	Hypo risk without CGM (HR+ACC+tremor probabilistic)	ROADMAP
SEN-ALG-010	Risk ladder L0–L3 deterministic rules	DESIGNED/ROADMAP
SEN-ALG-011	Medical-grade vs consumer-grade classification	ROADMAP
SEN-ALG-012	Sensor Correlation (Understand pillar)	ROADMAP
SEN-ALG-013	Duplicate detection hash for sensor sync	ROADMAP

19. MODULE REGISTER (18)

ID	Name	Maturity
SEN-MOD-001	sensor_record backend module	APPROVED_BACKEND_AUTHORITY

ID	Name	Maturity
SEN-M0D-002	sensor_timeline backend module	APPROVED_BACKEND_AUTHORITY
SEN-M0D-003	sensor_validation backend module	APPROVED_BACKEND_AUTHORITY
SEN-M0D-004	sensor_reliability_layer	APPROVED_BACKEND_AUTHORITY
SEN-M0D-005	sensor_conflict_detection	APPROVED_BACKEND_AUTHORITY
SEN-M0D-006	sensor_confidence_layer	APPROVED_BACKEND_AUTHORITY
SEN-M0D-007	Sensor Card UI component	DOCUMENTED
SEN-M0D-008	DeviceProfile / DeviceSession concepts	DESIGNED
SEN-M0D-009	DBL_TREMOR_V1 / DBL_BALANCE_V1 / DBL_RESP_PPG_V1 / DBL_POSTURE_V1	ROADMAP
SEN-M0D-010	DB9_HYPO_RISK_V1 / DB9_HYPO_ALERT_V1	ROADMAP
SEN-M0D-011	EE_EVENT_RELAY_V1 / EE_ESCALATE_V1 / EE_TELEPHONY_BRIDGE_V1	ROADMAP
SEN-M0D-012	Blind/Low-Vision haptic+voice alert mode	ROADMAP
SEN-M0D-013	Health Connect Platform V1 module	ROADMAP
SEN-M0D-014	Apple Health Platform V1 module	ROADMAP
SEN-M0D-015	Samsung Health Platform V1 module	ROADMAP
SEN-M0D-016	CGM Platform V1 module	ROADMAP
SEN-M0D-017	Blood Pressure Devices Platform V1 module	ROADMAP
SEN-M0D-018	sensorReadiness profile/UI hook	CODE_EXPORT

20. INGESTION REGISTER (14)

ID	Name / path	Maturity
ING-001	Band→BLE→Mobile Gateway→TLS→Cloud	DESIGNED
ING-002	/vX/devices/pair	DESIGNED
ING-003	/vX/devices/telemetry	DESIGNED
ING-004	/vX/devices/test/start finish	DESIGNED
ING-005	USAL ingress (all devices)	DESIGNED; OUT_OF_SCOPE real ingestion
ING-006	BSB canonical event pipeline	DESIGNED; OUT_OF_SCOPE brokers/queues
ING-007	/vX/biomarkers/evaluate	ROADMAP
ING-008	HomeLab webhook/API import	ROADMAP
ING-009	Mobile offline buffer (encrypted)	DESIGNED (policy TBD duration)
ING-010	Health Connect ingestion bridge	ROADMAP
ING-011	Apple Health ingestion bridge	ROADMAP
ING-012	Samsung Health ingestion bridge	ROADMAP
ING-013	Multi-wearable vendor bridges (Watch/Ring/WearOS/Fitbit/Garmin/Oura/WHOOP)	ROADMAP
ING-014	CGM / BP / Medical Devices platform ingestion	ROADMAP

Modes may include event/stream/batch/manual where documented. Offline buffer appears as designed path; retry/resilience details remain partial.

21. NORMALISATION REGISTER (9)

Normalisation ≠ clinical interpretation.

ID / entry	Summary
NORM-001	USAL canonical biological signal types

ID / entry	Summary
NORM-002	Device-specific schema containment at USAL boundary
NORM-003	Telemetry schema HR/accel/temp with UTC timestamps/units/quality flags
NORM-004	DeviceProfile / capabilities model (elisa_band_v1)
NORM-005	EE_EVENT_RELAY normalisation (risk_type, severity, timestamp, confidence)
NORM-006	Universal Sensor Record model (contract-cited)
NORM-007	measurement_grade normalisation field
NORM-008	measurement_timestamp vs received_timestamp + timezone sync model
NORM-009	duplicate_detection_hash for sync sessions

Includes where supported: unit/time/timezone/device normalisation, canonical schemas, adapters, duplicate handling, measurement_grade fields.

22. QUALITY / CONFIDENCE REGISTER (15)

First-class controls. Includes signal quality, confidence, reliability, freshness/stale handling, missing/noise/dropout/outliers, calibration/baseline where named, duplicate detection, and medical-grade vs consumer-grade distinction.

ID / entry	Summary
QC-001	signal_quality / motion_artifact / battery_low flags
QC-002	signal_confidence field on BSB events
QC-003	device_trust_score + integrity_checks (no implicit trust)
QC-004	Uncertain signals carry reduced confidence; unverified must not trigger sensitive logic
QC-005	Sensor validation detail projection
QC-006	Sensor reliability layer projection
QC-007	Sensor conflict detection projection
QC-008	Sensor confidence layer projection
QC-009	UI states: loading/available/stale/conflict/unverified/restricted/error
QC-010	Low-perfusion index flags (resp PPG)
QC-011	Feature flags OFF-by-default until evidence (DBL_*)
QC-012	Rate-limit + anomaly detection for telemetry
QC-013	Medical-grade vs consumer-grade classification control
QC-014	Automatic Sensor Validation control
QC-015	Medical Data Trust confidence/verification/clinical reliability fields

23. FUSION ENGINE REGISTER (6)

Do not call ordinary aggregation “sensor fusion” unless sources do. **FUSION ≠ DIAGNOSIS.**

ID / entry	Summary
FUS-001	DBL Core Engine (multi-signal → BiomarkerResult)
FUS-002	DB9_HYPO_RISK_V1 (CGM + HR drop + ACC tremor micro-signature)
FUS-003	BCE multi-dimension context enrichment
FUS-004	BDT multi-state aggregation
FUS-005	EE multi-signal escalation rules (tremor+HR, hypo, collapse)
FUS-006	Sensor Correlation (events↔context Understand pillar)

24. CROSS-PHASE OWNERSHIP MATRIX (18)

Never treat **REFERENCED BY PHASE** as **OWNED BY PHASE**.

ID	Relationship	Flow / role class	Maturity
XP-01	USAL/BSB → Phase 3/7/8/9 modules	normalised signals/context	DESIGNED
XP-02	DBL → Phase 8 AHE	biomarker early-warning	ROADMAP
XP-03	DBL/Phase9 → EEE	risk events	ROADMAP
XP-04	EEE → Phase 3 SOS	emergency escalation adjacency	ROADMAP
XP-05	CTD → Phase 8 Passport/Identity	role permissions	ROADMAP
XP-06	MJ Sensor UI → Phase 1 Permissions / Phase 9 audit / Phase 11 family	gates	DOCUMENTED_BINDING
XP-07	HomeLab → AHE/Diabetes	lab results	ROADMAP
XP-08	Sensor Showcase → future Elisence hardware	compatibility	DESIGNED
XP-09	BDT hormonal state → Phase 7 Women+	state read	DESIGNED
XP-10	Glucose/hypo → Phase 9 Diabetes Core	domain interpretation	ROADMAP
XP-11	Care messages → Phase 13 Dopamine (future mention)	future non-dependent link	FUTURE
XP-12	Competitive gap doc → Phase 8/9 capability proposals	strategy	DOCUMENTED
XP-013	MJ Phase4 measurements sensor-ready but not sensor implementation → ADJACENT/PRECONDITION	ADJACENT/PRECONDITION	ROADMAP/DOCUMENTED
XP-014	MJ Phase5 Sensor & Wearable Platform programme ownership lineage → SOURCE/INGESTION/NORMALISER (roadmap)	SOURCE/INGESTION/NORMALISER (roadmap)	ROADMAP/DOCUMENTED
XP-015	MJ Phase6 Sensor Intelligence consumer → INTERPRETER/CONSUMER	INTERPRETER/CONSUMER	ROADMAP/DOCUMENTED
XP-016	Medication Phase2 ↔ Sensor Data / Elisa Phase5 reminders → CONSUMER/CONTEXT	CONSUMER/CONTEXT	ROADMAP/DOCUMENTED
XP-017	Family Phase11K Remote Monitoring consumer → CONSUMER	CONSUMER	ROADMAP/DOCUMENTED
XP-018	Phase5 numbering collision: Sensor Platform vs ELISA companion → CONFLICT/CROSS_PHASE_EVIDENCE	CONFLICT/CROSS_PHASE_EVIDENCE	ROADMAP/DOCUMENTED

Role vocabulary for deeper reading of archaeology matrices: SOURCE, ACQUISITION, INGESTION, NORMALISER, QUALITY OWNER, INTERPRETER, DOMAIN CONSUMER, PROJECTOR, ELISA CONSUMER, ALERT/PATHWAY OWNER, ADJACENT, UNKNOWN, CROSS_PHASE_OWNER.

25. DOMAIN CONSUMER REGISTER (10)

Domain	Purpose	Maturity
AHE / Phase 8 Blood Intelligence	early warning / blood intelligence	DOCUMENTED_CONSUMER_DESIGN
Diabetes / Phase 9	diabetes monitoring + hypo escalation	ROADMAP
Women+ / Phase 7	women health context consumption	DESIGNED_SUBSCRIBER
Body Intelligence / Phase 3	body intelligence + SOS	DESIGNED_SUBSCRIBER
Care Team / Guardians	non-diagnostic support	ROADMAP
Governments / Clinics (aggregate)	population transparency	CONCEPT/ROADMAP
Medical Journey UI Sensor surfaces	observational sensor UI	APPROVED_BACKEND_AUTHORITY
Medication adherence (sensor as context only)	Medication adherence (sensor as context only)	ROADMAP/DOCUMENTED
Family/Care remote monitoring	Family/Care remote monitoring	ROADMAP/DOCUMENTED
Medical Journey Timeline / Trust Layer (sensor-origin events)	Medical Journey Timeline / Trust Layer (sensor-origin events)	ROADMAP/DOCUMENTED

Domain intelligence owner ≠ sensor owner. Domain may consume raw, derived, or contextual signals.

26. ELISA ↔ SENSOR RELATIONSHIPS (5)

Critical law:

- ELISA CONSUMES SIGNAL ≠ ELISA OWNS SENSOR
- ELISA CONSUMES DOMAIN OUTPUT ≠ ELISA OWNS DOMAIN INTELLIGENCE

ID	Relationship	Type	Note
ELISA-SEN-001	Elisa Band hardware naming vs ELISA companion layer	NAMING_ADJACENCY	Corpus uses “Elisa Band” for \$12 wearable; does NOT prove ELISA Phase-5 companion owns sensors.
ELISA-SEN-002	Voice/haptic alerts on phone for blind mode	INTERACTION_SURFACE	Voice instructions/TTS emergency copy; platform speech ≠ ELISA intelligence ownership.
ELISA-SEN-003	Cross-phase companion consumption of AHE/Diabetes signals (external Phase5 archaeology)	EXTERNAL_CROSS_REF_ONLY	Not evidenced inside this 17-file Sensor corpus; do not import Phase5 findings as Sensor-corpus proof.
ELISA-SEN-004	Elisa Phase5 medication reminders/voice with sensor-ready context	INDIRECT/CONTEXT	ELISA CONSUMES context ≠ owns sensor platform
ELISA-SEN-005	Elisa Band naming as hardware feeder (not companion ownership proof)	HARDWARE_NAMING_ADJACENT	

27. SENSOR → CHART RELATIONSHIPS (6)

Not full Chart archaeology — relationship map only.

ID	Signal / theme	Chart / visualisation	Maturity
SCH-001	BiomarkerScores / glucose/weight trends	Care Team Dashboard trend views	ROADMAP
SCH-002	RiskEvents L1-L3	Unified Risk Timeline (user + clinical/gov aggregate)	ROADMAP
SCH-003	Sensor records	Sensor Timeline / Sensor Card UI	DOCUMENTED_BINDING
SCH-004	Abstract public indicators	Sensor Showcase public layer indicators	DESIGNED
SCH-005	SedentaryRiskScore / activity	AHE/Diabetes dashboard trends (non-disease labels)	ROADMAP
CHART-SEN-006	Doctor-ready summaries/charts consuming sensor-aware intelligence	Doctor-ready summaries/charts consuming sensor-aware intelligence	ROADMAP

28. PRIVACY / PERMISSION / SAFETY / AUDIT / RESILIENCE

Baseline-preserved register sizes (V2 did not renumber these):

Register	Count
Privacy / permission controls	10
Sensor safety controls	12
Provenance / audit controls	7
Resilience controls	5
Data stores / registries	5
Routes / endpoints	6

Safety themes preserved: non-diagnostic posture; do not act on stale/noisy signals without thresholds; consumer vs medical-device boundary; escalation confirmation; seek-care language; low-confidence suppression; fail-closed where documented.

Privacy themes preserved: purpose limitation, consent layers, wearable/Health-store permissions as design, minimisation, revocation, role access — without inventing retention periods ([SENSOR-G-08](#) UNKNOWN).

29. TEST / SEAL / DEVICE-PROOF

Register	Count	Golden-safe status
Test families	8	MIXED documentary claims; raw logs ABSENT
Device-proof items	4	No hardware device-proven completion in corpus

Do **not** say “Sensor was not tested.” Say: mixed planned/acceptance language exists; raw pass packs absent; confidential “300+ PASS” claims are **CONFLICTED** with archaeology ([SENSOR-C-10](#)).

Route exists ≠ device proof. Binding approved ≠ band manufactured.

30. HISTORICAL ERAS (6 + late-source extensions)

SEN-ERA-1 — Competitive Gap Framing (Sensor-Agnostic Intent)

- MATURITY: [DOCUMENTED](#)
- NOTES: Phase9-sensor.pdf compares Huma/Livongo/Omada; proposes Digital Biomarker Lab / Emergency Escalation / Care Team; Sensor-Agnostic claim for Phase 9.
- SOURCES: ["[SEN-SRC-012](#) ", "[SEN-SRC-013](#) "]

SEN-ERA-2 — Elisa Band + Digital Biomarker Master Roadmap

- MATURITY: [DESIGNED](#) / [NOT_DONE](#)
- NOTES: Roadm map. Sensor Phase9.pdf — Design Only; Elisa Band v1 HR/ACC/optional temp; DBL/EEE/CTD waves; explicit Everything=NOT DONE until evidenced.
- SOURCES: ["[SEN-SRC-014](#) ", "[SEN-SRC-015](#) "]

SEN-ERA-3 — DBL / Emergency Add-On Extensions

- MATURITY: [ROADMAP](#)
- NOTES: Add-Roadmap-Sensor + What sensors does — tremor/hypo/respiratory/posture add-ons integrating Phase 8+9 without new hardware.
- SOURCES: ["[SEN-SRC-009](#) ", "[SEN-SRC-017](#) "]

SEN-ERA-4 — Strategic / Investor Sensor Narrative

- MATURITY: [CONCEPT](#) / [STRATEGIC](#)
- NOTES: Sensors layer document + Disney strategic statement; \$12 sensor narrative; claims design complete / development pending — not runtime proof.
- SOURCES: ["[SEN-SRC-016](#) ", "[SEN-SRC-010](#) "]

SEN-ERA-5 — Phase 10 USAL / BSB / BCE / BDT Infrastructure Architecture

- MATURITY: [DESIGNED](#) (architecture-only; explicit OUT OF SCOPE for ingestion/drivers)
- NOTES: Phase10-sensor.pdf sections 2.3–2.7 define Sensor Showcase, USAL, Biological Signal Bus, Biological Context Engine, Biological Digital Twin with multi-phase subscribers.
- SOURCES: ["[SEN-SRC-011](#) "]

SEN-ERA-6 — Medical Journey UI Sensor Record Authority Bindings

- MATURITY: [APPROVED_BACKEND_AUTHORITY](#) (UI binding; not device-proof)
- NOTES: Sensor Card + backend bindings for sensor_record/timeline/validation/reliability/conflict/confidence; Phase1 permissions + Phase9 audit + Trust Layer dependencies.
- SOURCES: ["[SEN-SRC-001](#) ", "[SEN-SRC-002](#) ", "[SEN-SRC-003](#) ", "[SEN-SRC-004](#) ", "[SEN-SRC-005](#) ", "[SEN-SRC-006](#) ", "[SEN-SRC-007](#) ", "[SEN-SRC-008](#) "]

Late-source era extensions (not renumbering baseline six)

- Master Product Roadmap v1.1 sensor-aware MJ programme ([SEN-VF-009](#))
- Strategic expansion DBL/EEE/CTD in-progress crown statement ([SEN-VF-010](#))
- Internal hardware economics / OEM due-diligence brief ([SEN-VF-011](#))
- Composite Phase10/social PRD dump with Devices & Sensors settings ([SEN-VF-007](#))

- Premium UI code export with sensorReadiness (`SEN-VF-008`)

31. CONFLICT REGISTER (11)

`SENSOR-C-01` — disposition `LEAVE_UNCHANGED`

- NOTE: Reinforced: Persian says design complete / implementation starting; confidential claims operational

`SENSOR-C-02` — disposition `LEAVE_UNCHANGED`

- NOTE: Master Roadmap still lists CGM/medical devices alongside low-cost band strategy

`SENSOR-C-03` — disposition `PARTIALLY_RESOLVE`

- NOTE: Design intent now explicit Apple Health/Health Connect programme; implementation still unproven

`SENSOR-C-04` — disposition `CREATE_STRENGTHENED_VIA_C-08`

- NOTE: Phase ownership still cross-cutting; numbering collision added

`SENSOR-C-05` — disposition `PARTIALLY_RESOLVE`

- NOTE: Master Roadmap separates Elisa Phase5 companion duties from Phase5 Sensor Platform numbering—but creates lineage conflict

`SENSOR-C-06` — disposition `LEAVE_UNCHANGED`

`SENSOR-C-07` — disposition `LEAVE_UNCHANGED`

`SENSOR-C-08` — disposition `CREATE`

- TITLE: `SENSOR_PHASE_NUMBERING_LINEAGE_CONFLICT`
- QUESTION: Is Phase 5 Sensor & Wearable Platform or ELISA?
- SOURCE A: Sensor-Main Master Roadmap v1.1: Phase 5 = Sensor & Wearable Platform
- SOURCE B: Established Phase5 ELISA archaeology: Phase 5 = ELISA
- SAFE INTERPRETATION: Treat Sensor as cross-cutting; do not rely on a single phase number
- CLASS: `DIFFERENT_ROADMAP_ERA` / `PARALLEL_SUBSYSTEM_NUMBERING` / `DOCUMENT_VERSION_DRIFT` (not silently resolved)

`SENSOR-C-09` — disposition `CREATE`

- TITLE: Elisa Band price \$12 vs Gen1 \$21
- SOURCE A: Confidential + Persian \$12
- SOURCE B: Persian Elisa Band Gen1 \$21
- SAFE INTERPRETATION: Price not locked; `PRODUCT_CONCEPT` only

`SENSOR-C-10` — disposition `CREATE`

- TITLE: Due-diligence operational/test claims vs archaeology NOT DONE
- SOURCE A: Confidential: 300+ tests PASS, engines operational
- SOURCE B: Prior Sensor archaeology: raw logs ABSENT; Design Only
- SAFE INTERPRETATION: `INTERNAL_COMMERCIAL_CLAIM_ONLY` / `CONFLICTED`

`SENSOR-C-11` — disposition `CREATE`

- TITLE: Premium UI naming vs Premium sensor hardware tier
- SOURCE A: Sensor-2 Premium* SwiftUI components + sensorReadiness
- SOURCE B: No STANDARD/PREMIUM hardware feature split locked
- SAFE INTERPRETATION: `STANDARD_PREMIUM_FINAL_FEATURE_SPLIT` =NOT_DOCUMENTARILY_LOCKED

32. GAP REGISTER (14)

SENSOR-G-01 — disposition LEAVE_UNCHANGED

SENSOR-G-02 — disposition LEAVE_UNCHANGED

- NOTE: Economics doc adds claims but not hardware proof

SENSOR-G-03 — disposition PARTIALLY_RESOLVE

- NOTE: Named platform programme now explicit in Master Roadmap; implementation artefacts still absent

SENSOR-G-04 — disposition LEAVE_UNCHANGED

- NOTE: Confidential test claims remain unverified; no raw logs added

SENSOR-G-05 — disposition LEAVE_UNCHANGED

SENSOR-G-06 — disposition LEAVE_UNCHANGED

SENSOR-G-07 — disposition LEAVE_UNCHANGED

SENSOR-G-08 — disposition LEAVE_UNCHANGED

SENSOR-G-09 — disposition PARTIALLY_RESOLVE

- NOTE: Master Roadmap links Elisa Phase5 to sensor-ready medication context; still not ELISA owns sensor

SENSOR-G-10 — disposition LEAVE_UNCHANGED

SENSOR-G-11 — disposition CREATE

- MISSING: China/OEM manufacturer identity, contacts, quotations, MOQ, tooling, sample orders
- WHY IT MATTERS: Industrialisation maturity cannot advance beyond FOUNDER_HISTORY_ONLY / commercial claim

SENSOR-G-12 — disposition CREATE

- MISSING: STANDARD vs PREMIUM hardware feature split lock
- WHY IT MATTERS: No documentary lock of tier differentiation

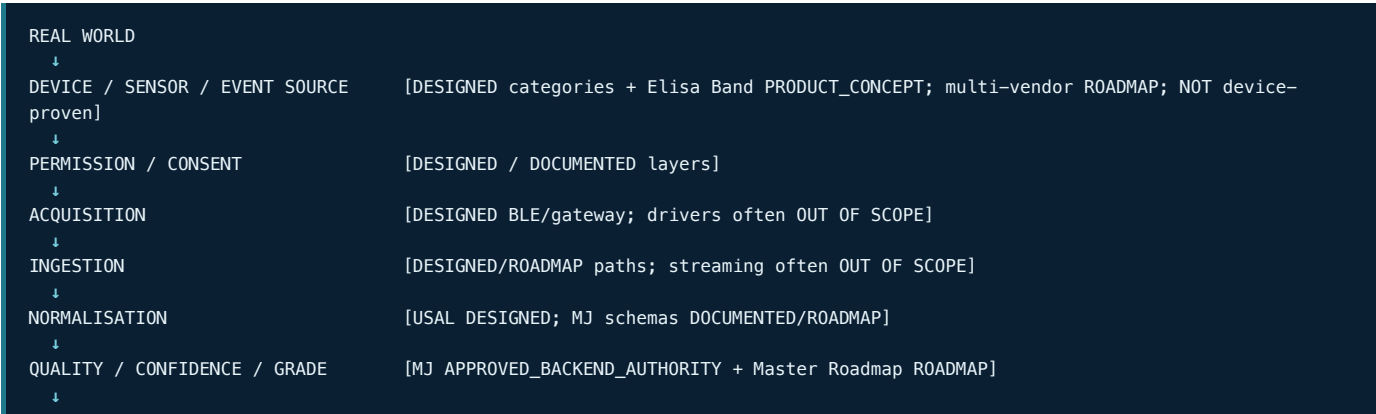
SENSOR-G-13 — disposition CREATE

- MISSING: Independent corroboration of due-diligence unit economics and scale claims
- WHY IT MATTERS: DOCUMENT_STATES ≠ externally verified

SENSOR-G-14 — disposition CREATE

- MISSING: Dedicated OEM engagement evidence trail (dates/messages/quotes)
- WHY IT MATTERS: Filename implies factory doc; body is investor economics brief without factory identity

33. ZERO-LOSS END-TO-END ARCHITECTURE MODEL



PROVENANCE / MEDICAL DATA TRUST	[DOCUMENTED/DESIGNED]
↓	
DOMAIN INTERPRETATION	[DBL/AHE/Diabetes/Women+/Medication context – ROADMAP/DESIGNED; non-diagnostic]
↓	
SAFETY / POLICY	[DESIGNED/ROADMAP ladders; fail-safe language]
↓	
ELISA / HUMAN INTERACTION	[INDIRECT/CONTEXT; does not own sensor plane]
↓	
CHART / PROJECTION	[DOCUMENTED_BINDING / ROADMAP]
↓	
PATHWAY / HUMAN DECISION	[ROADMAP / guardian / care pathways]

Full plane runtime proof: **UNPROVEN**.

34. GOLDEN CLAIM BOUNDARIES (SUMMARY)

Safe to say	Not safe to say
Sensor/signal is cross-cutting infrastructure by design	Sensor is permanently Phase 5
Design intent for multi-domain signal ecosystem is SUPPORTED	Full runtime plane is proven
Architecture is sensor-agnostic by design	Health Connect/Apple Health bridges are live
MJ sensor bindings are APPROVED_BACKEND_AUTHORITY	Bindings equal device-proven band telemetry
Affordable ~\$12 screenless band was a product strategy	\$12 band equals Apple Watch clinically
OEM licensing was commercially discussed in due-diligence doc	China factory engagement is proven in this corpus
Sensor may support medication context	Sensor proves medication taken
ELISA may consume signal/context	ELISA owns Sensor
Fusion/correlation may exist as designed	Fusion equals diagnosis
Mixed test claims exist	300+ PASS is independently proven here

35. AUTHORITY CHAIN FOR FUTURE WORK

1. This Golden (reconciled V2)
2. 10_SENSOR_RECONCILED_CENSUS_LOCK_V2.json
3. Late-source reconciliation report V1
4. Baseline 17-file archaeology report (historical; incomplete alone)
5. Canonical sources under /SENSOR/sources (22 files)

PDF generation is a **separate mission**.

36. IMMUTABILITY STAMP

- SOURCE_FILES_MODIFIED=0
- SOURCE_FILES_DELETED=0
- CODE_FILES_MODIFIED=0
- TESTS_EXECUTED=NO
- GIT_MUTATIONS=0
- DEPLOYMENT_PERFORMED=NO
- NEW_ARCHAEOLOGY=NO
- NEW_SOURCE_ACQUISITION=NO
- GOLDEN_PDF_CREATED=NO
- GOLDEN_CREATED=YES

END OF GOLDEN REFERENCE